

Euclidean and Non-Euclidean Geometry:

Projects:

1. (pg 139) "Claim from Example 3 - Euclidean Plane"

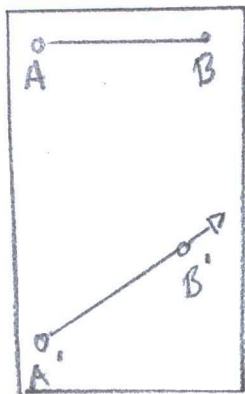
"For two segments, same distance between points is congruent."

"For two angles, same angle measure is congruent."

(pg 139) "Distance Formula"

$$d(AB) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2} \quad \text{with points}$$
$$A = (a_1, a_2)$$
$$B = (b_1, b_2)$$

Claims hold for Axioms C-1, 2, 3, 4, 5:



(pg 119) "Congruence Axiom #1"

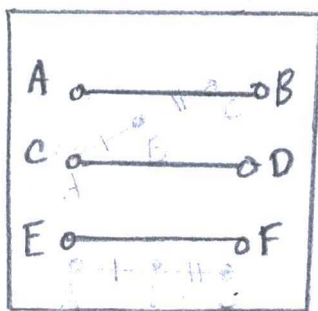
Proof

Hypothesis

Evidence

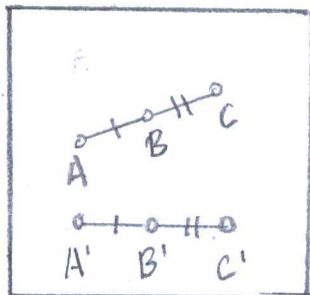
$$1) AB \cong A'B'$$

$$d(AB) = d(A'B')$$



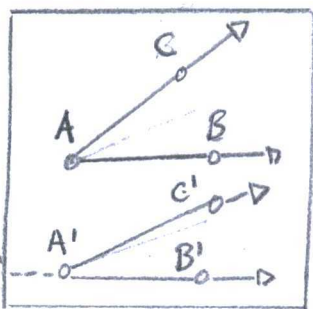
(pg 120) "Congruence Axiom #2"

Proof:	Hypothesis	Evidence
1)	$AB \cong CD$	$d(AB) \cong d(CD)$
2)	$AB \cong EF$	$d(AB) \cong d(EF)$
3)	$CD \cong EF$	$(AB \cong CD) \& (AB \cong EF) \rightarrow CD \cong EF$



(pg 120) "Congruence Axiom #3"

Proof:	Hypothesis	Evidence
1)	$AB \cong A'B'$	$d(AB) \cong d(A'B')$
2)	$BC \cong B'C'$	$d(BC) \cong d(B'C')$
3)	$AC \cong A'C'$	Proposition 3.19



(pg 120) "Congruence Axiom #4"

Proof:	Hypothesis	Evidence
1)	$BC \cong B'C'$	$d(BC) \cong d(B'C')$
2)	$\angle BAC \cong \angle B'A'C'$	$(BC \cong B'C') \rightarrow \angle BAC \cong \angle B'A'C'$

(pg 121) "Congruence Axiom #5"

Proof:	Hypothesis	Evidence
1)	$\angle A \cong \angle B$	Axiom C-4
2)	$\angle A \cong \angle C$	Axiom C-4
3)	$\angle B \cong \angle C$	$(\angle A \cong \angle B) \& (\angle A \cong \angle C) \rightarrow \angle B \cong \angle C$

2.

Citation: "Algebraic Geometry"

Hartshorne, Robin

University of California, USA

Exercise 6.D on page 148 describes

fields in ratios, subgroups, and rank.

$K(X)$ is "the Grothendieck Group"

and the quotient of the subgroup.

$K(X)$ is real when X is a ratio.

Concepts, such as "coherent sheaves"

define ratios between groups with

generic rank.

The congruence axioms hold when

ratios are real ($\mathbb{R}$) ratios!

Segments at $\sqrt{2}$ -lengths cannot

match the model by Hartshorne.

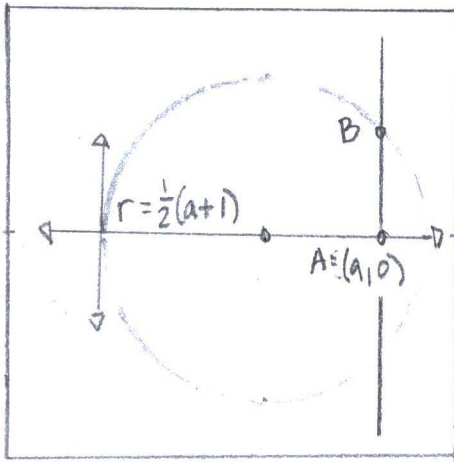
3. Section 17 not shown in same book above.

4. In essence, standard methods in geometry should apply to motion, Specifically, geometry requires rigorous verification throughout transformation, reflection, and rotation.

4. Citation: "Elementary geometry from an Advanced Standpoint."
Moise, Edwin.
Queens College of City University
of New York (1990).

Edwin ordered the lengths in a triangle for "The Triangle Theorem."
Rank had a benefit to similar side ranks in circle-circle continuity and Euclids Postulates.

5



σ = circle

r = radius

$$= \frac{1}{2}(a+1)$$

A = Point

$$= (a, 0)$$

B = Point intersecting $x=a$ and circle

Proof about $B = (a, \sqrt{a})$:

X-coordinate is a at $x=a$

Y-coordinate relates to Pythagoreous'

Law. The base is $a - \frac{1}{2}(a+1)$,

hypotenuse $\frac{1}{2}(a+1)$, so the leg

and y-coordinate $y = \sqrt{a}$.

$$B = (a, \sqrt{a})$$

6.

End of Chapter.